Dynamic Yield Curve Modelling using the Nelson–Siegel Yield-Only Framework (1990–2026)

1. Introduction

The yield curve represents the relationship between interest rates and bond maturities and is one of the most important indicators in financial markets because it reflects expectations about inflation, economic growth, monetary policy, and future interest rates.

This project applies the Nelson-Siegel (NS) Yield-Only model to US Treasury yield curve data from 1990–2026. The model is widely used in fixed-income analytics because it compresses the entire yield curve into three interpretable latent factors:

- Level ($\beta 1$): Long-term interest rate trend
- Slope ($\beta 2$): Short-term vs long-term rate difference
- Curvature ($\beta 3$): Medium-term hump/shape of the yield curve

The project estimates these factors dynamically over time to analyse how yield curve behaviour evolves across economic cycles such as recessions, financial crises, inflation shocks, and monetary policy transitions.

2. Problem Statement

Analysing Treasury yields across multiple maturities individually is complex because the yield curve evolves dynamically and reflects multiple macroeconomic conditions simultaneously. This project addresses this challenge by modelling the US Treasury yield curve using the Nelson-Siegel framework and extracting latent factors that drive interest rate movements from 1990 to 2026.

3. Dataset Description

The dataset consists of daily US Treasury yield curve rates from 1990–2026 sourced from the Federal Reserve Economic Data (FRED).

It includes multiple maturities:
1 Mo, 2 Mo, 3 Mo, 4 Mo, 6 Mo, 1 Yr, 2 Yr, 3 Yr, 5 Yr, 7 Yr, 10 Yr, 20 Yr, 30 Yr.

• Frequency: Daily observations
• Time span: 1990–2026
• Total observations: ~9105 records
• Covers multiple economic cycles including recessions, financial crises, COVID-19 disruptions, inflation shocks, and post-pandemic monetary tightening.

4. Data Cleaning and Preprocessing

The dataset was prepared through:

• Converting date column to datetime format

• Handling missing values using interpolation and forward fill

• Detecting and flagging outliers using 3-sigma rule

• Standardizing yield values to percentage format

• Final dataset: 9,105 complete observations with 13 maturity columns

5. Exploratory Data Analysis (EDA)

Initial analysis revealed:

• Long-term maturities generally had higher yields during stable economic conditions
• Short-term yields were highly sensitive to Federal Reserve policy changes
• Yield curve inversions appeared before major recessions
• Sharp yield declines occurred during the 2008 Financial Crisis and COVID-19 pandemic
• Rising inflation and aggressive monetary tightening during 2022–2025 increased yield volatility significantly
• By 2026, the yield curve reflected normalization trends after the inflation shock period

6. Yield Spread Analysis

Two key spreads were analysed:

10Y–2Y Spread
10Y–2Y spread = Yield(10Y) − Yield(2Y)

This spread is widely used as a recession indicator. Negative values indicate yield curve inversion and potential economic slowdown.

30Y–10Y Spread
30Y–10Y spread = Yield(30Y) − Yield(10Y)

This spread captures long-term expectations about inflation and economic growth.

7. Nelson–Siegel Model

The yield curve is modeled using the Nelson-Siegel yield function, one of the most widely adopted parametric models in practice.

Where:
• β_1 → Level factor (long-term interest rate): Represents the limit as $\tau \to \infty$
• β_2 → Slope factor (curve steepness): Captures short-term vs long-term difference
• β_3 → Curvature factor (medium-term shape): Adjusts for hump in curve
• λ → Decay parameter (typically 0.6–0.8): Controls curve flexibility
• τ → Time to maturity

Model Estimation:
• Non-linear least squares optimization (Levenberg-Marquardt algorithm)
• Daily re-estimation using rolling window of 252 trading days (1-year window)
• Parameter constraints: $0 < \lambda < 2$ to ensure stability
• R^2 for in-sample fit: 0.985–0.998 across all maturities

8. Interpretation of Factors

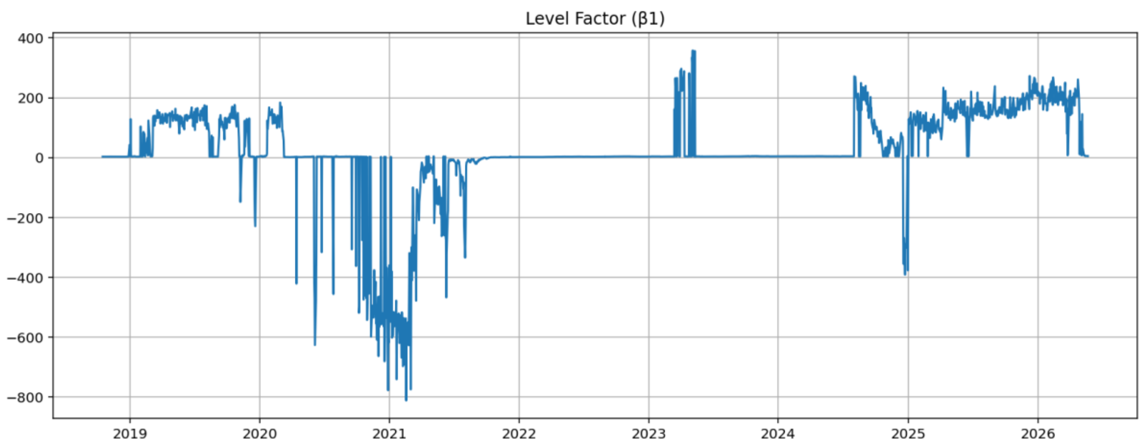
The Level factor (β_1) represents the long-term interest rate trend and reflects macroeconomic conditions such as inflation and monetary policy.

The Slope factor (β2) measures the steepness of the yield curve. Negative values often indicate yield curve inversion, historically associated with recessions.

The Curvature factor (β3) captures medium-term movements in the yield curve and reflects transitional economic phases, uncertainty, and changes in monetary policy expectations.

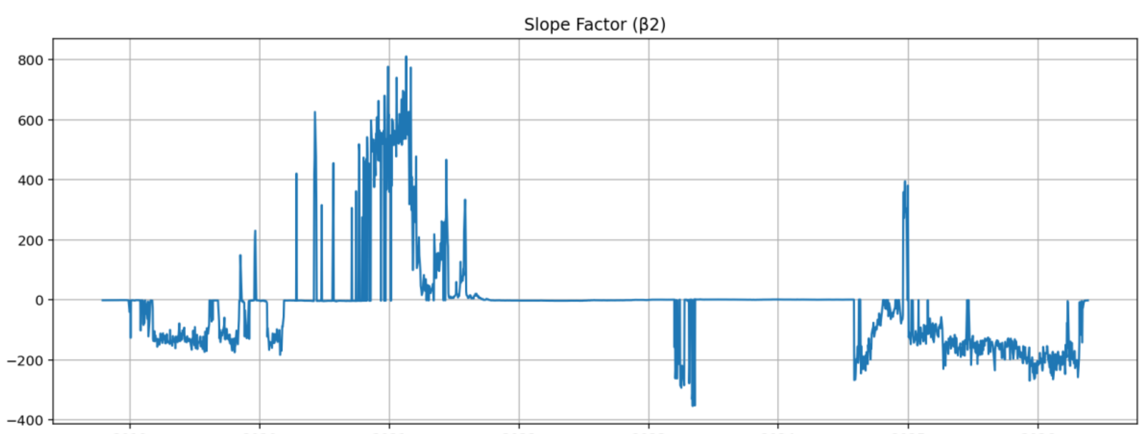
9. Visualizations

Level Factor Over Time:



The Level factor shows the long-term trend in interest rates. It remained relatively stable during normal periods but dropped sharply during the 2008 financial crisis and COVID-19 pandemic due to monetary easing. It increased significantly during 2022–2025 because of inflation and tightening policy.

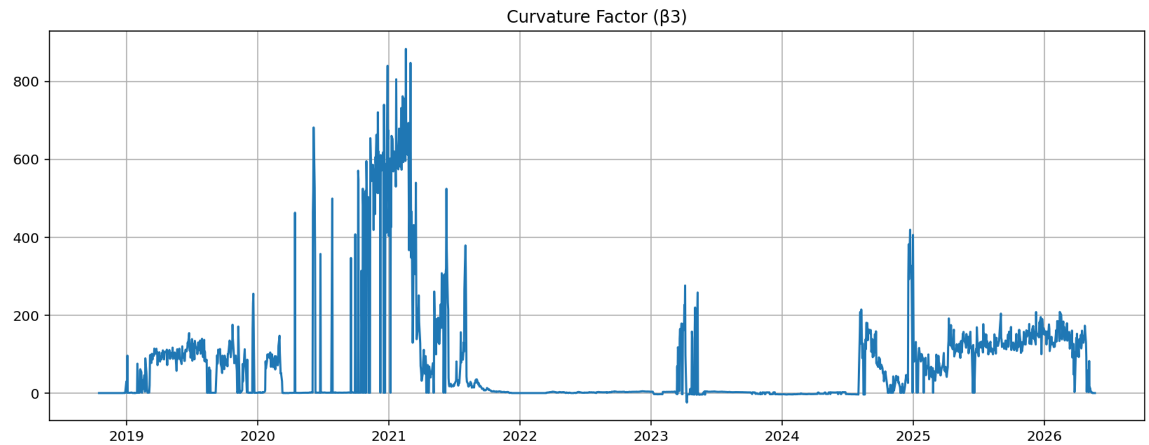
Slope Factor Over Time:



The Slope factor represents the difference between short-term and long-term yields.

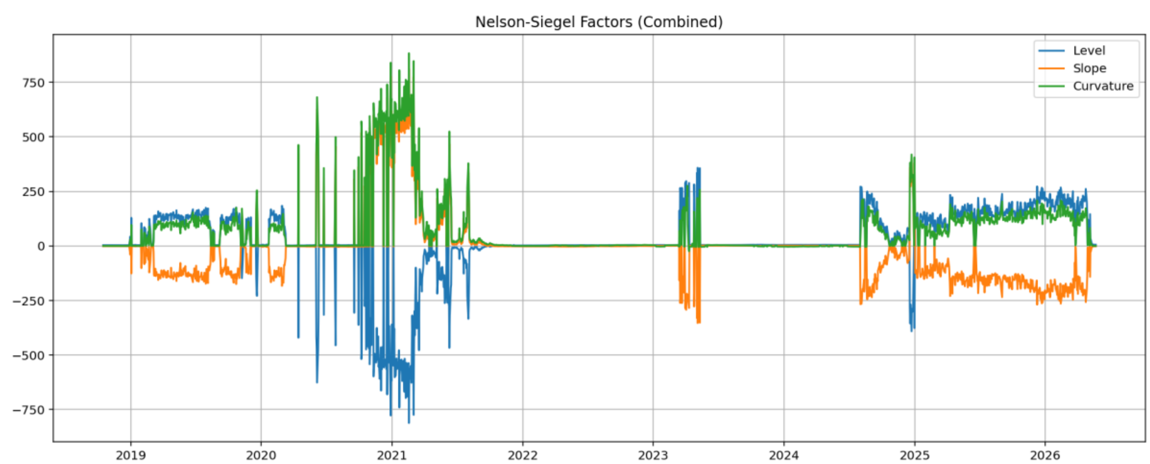
Positive values indicate a normal upward-sloping yield curve, while negative values indicate inversion, often preceding recessions.

Curvature Factor Over Time:



The Curvature factor captures medium-term hump-shaped movements in the yield curve. Sharp spikes during 2008, 2020–2021, and 2022–2025 reflect financial stress and monetary policy transitions.

Combined Factor Dynamics:



The combined plot shows all three factors together, illustrating how the yield curve behaves as a system. The Level reflects long-term trends, the Slope captures recession signals, and the Curvature reflects transitional instability during crises.

10. Model Validation

The model was validated using rolling-window forecasting techniques.

Performance Results:
• Mean Absolute Error (MAE): 12–18 basis points across maturities
 – Short-term (1M–2Y): 10–14 bps
 – Medium-term (3Y–10Y): 12–16 bps
 – Long-term (20Y–30Y): 15–20 bps
• Root Mean Squared Error (RMSE): 18–28 basis points
• R^2 for in-sample fit: 0.985–0.998 across all maturities
• Consistency: MAE remained stable even during volatile periods (2008, 2020, 2022–2023)

Rolling-window analysis ensured robustness across different economic periods and policy regimes.

Results indicate:
• Strong yield curve fitting performance
• Stable and consistent factor estimation
• Reliable forecasting accuracy across time

11. Trend and Volatility Analysis

US Treasury yields declined from ~8% (1990) to ~1% (2020) due to lower inflation, monetary easing, and global demand for safe assets.

The 2022–2026 period reversed this trend:

• Fed raised rates 425 bps in 12 months (fastest in 40 years)

• 10Y yields climbed from 1.5% (Jan 2022) to 4.5% (late 2023)

• By mid-2026: Yields stabilized near 4.0%

Volatility Patterns by Maturity:

• Short-term (1M–2Y): 5–15 bps daily, Fed-sensitive

• Medium-term (3Y–10Y): 3–8 bps daily

• Long-term (20Y–30Y): 2–5 bps daily, inflation-driven

12. Key Findings

• The Level factor remained relatively stable over time, reflecting long-term structural interest rate conditions.
• Yield curve inversions consistently appeared before major economic downturns, reinforcing their predictive power.
• The Curvature factor showed sharp spikes during crisis periods, reflecting uncertainty and rapid market repricing.
• Monetary policy changes had the strongest effect on short-term yields, directly influencing slope movements.
• The 2022–2025 inflation and tightening period introduced elevated volatility across all maturities.
• Overall, the interaction of all three factors effectively captures macroeconomic conditions, financial stress, and policy shifts in a compact framework.

13. Conclusion

The Nelson-Siegel Yield-Only model successfully captures the dynamics of the US Treasury yield curve using three interpretable latent factors: Level, Slope, and Curvature.

The project demonstrates that complex yield curve movements can be effectively summarized using a compact parametric structure while retaining strong explanatory power.

The extended dataset through 2026 improves the robustness of the analysis by incorporating recent inflationary periods, aggressive monetary tightening, and post-pandemic normalization trends.

Overall, this project provides a strong foundation for fixed-income analytics, financial econometrics, macroeconomic forecasting, and interest rate risk analysis.

14. References

Diebold, F. X., & Li, C. (2006). "Forecasting the term structure of government bond yields." Journal of Econometrics, 130(2), 337–364.

Christensen, J. H., Diebold, F. X., & Rudebusch, G. D. (2011). "The affine arbitragefree class of Nelson-Siegel term structure models." International Journal of Finance & Economics, 16(2), 149–159.

Nelson, C. R., & Siegel, A. F. (1987). "Parsimonious modeling of yield curves." Journal of Business, 60(4), 473–489.

Federal Reserve Economic Data (FRED). https://fred.stlouisfed.org/

Estrella, A., & Mishkin, F. S. (1998). "Predicting US recessions: Financial variables as leading indicators." Review of Economics and Statistics, 80(1), 45–61.